In [1]:
```
#Name : Vaishnavi Kadam
#Roll no :9
```

In [2]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

In [3]:
```
data = sns.load_dataset("titanic")
data.head()
```

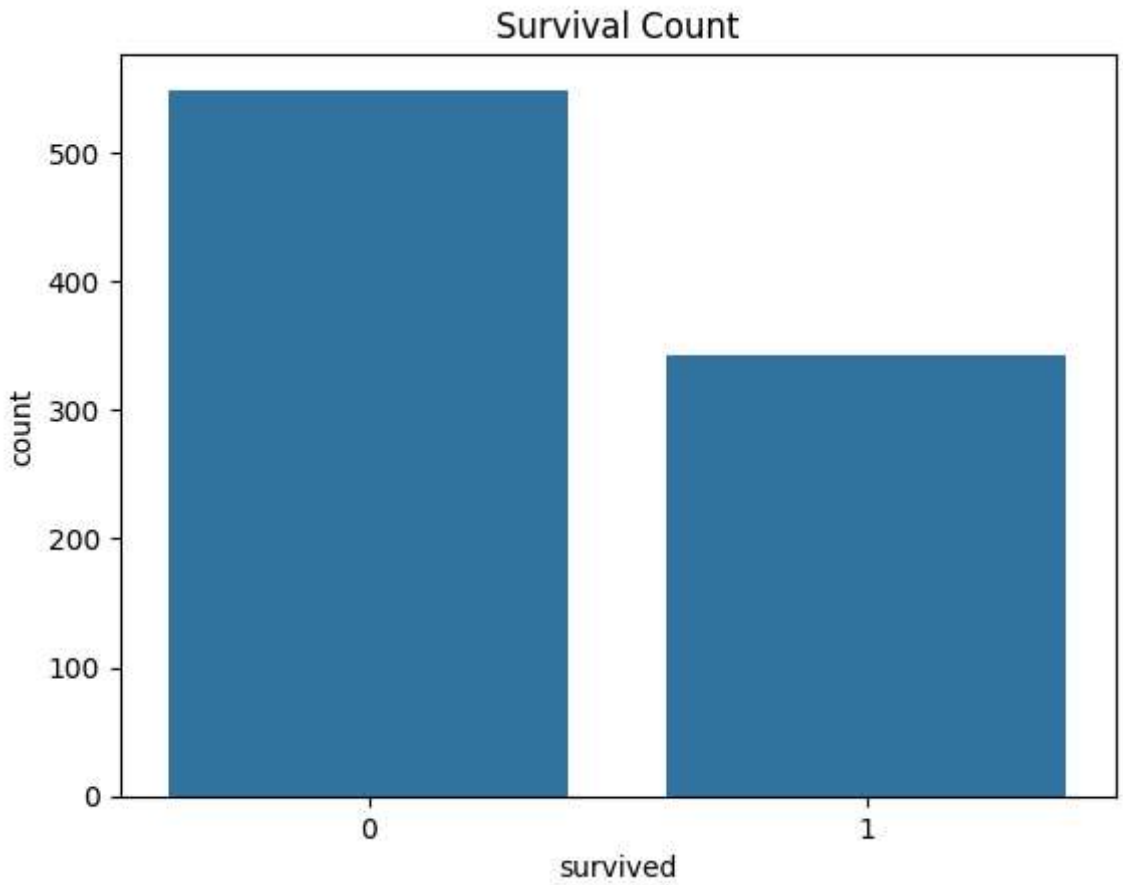
Out[3]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adul
0	0	3	male	22.0	1	0	7.2500	S	Third	man	
1	1	1	female	38.0	1	0	71.2833	C	First	woman	
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	
3	1	1	female	35.0	1	0	53.1000	S	First	woman	
4	0	3	male	35.0	0	0	8.0500	S	Third	man	

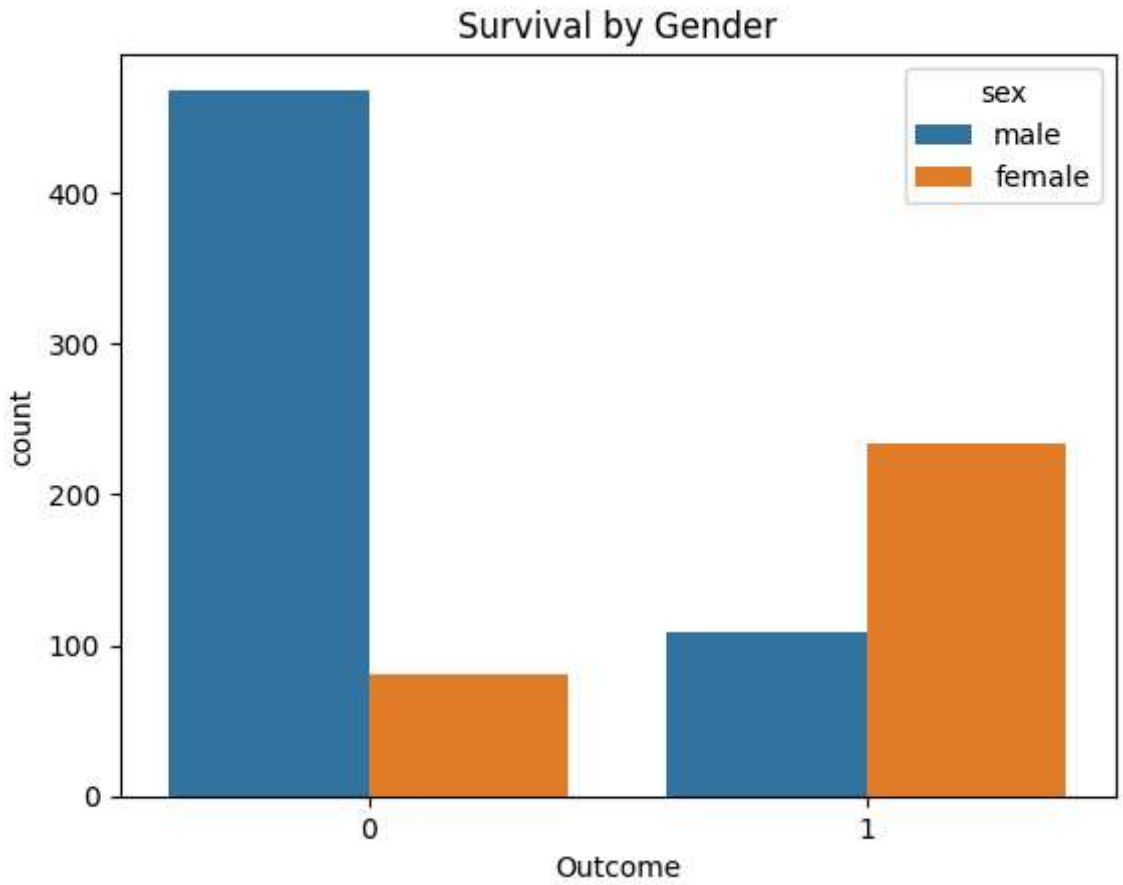
In [4]:
```
data.shape
```

Out[4]:
```
(891, 15)
```

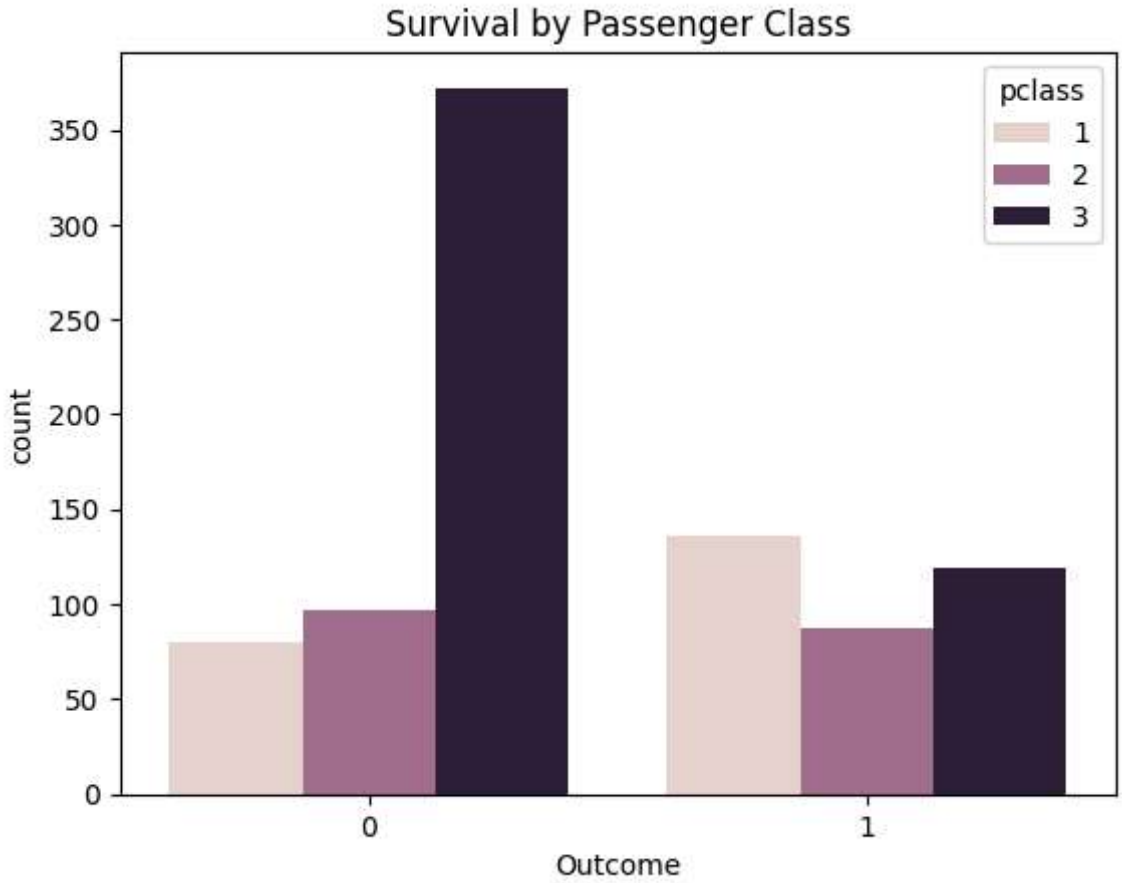
In [5]:
```
#Survival count
sns.countplot(x='survived', data=data)
plt.title("Survival Count")
plt.show()
# 0 is not survived and 1 is survived
```



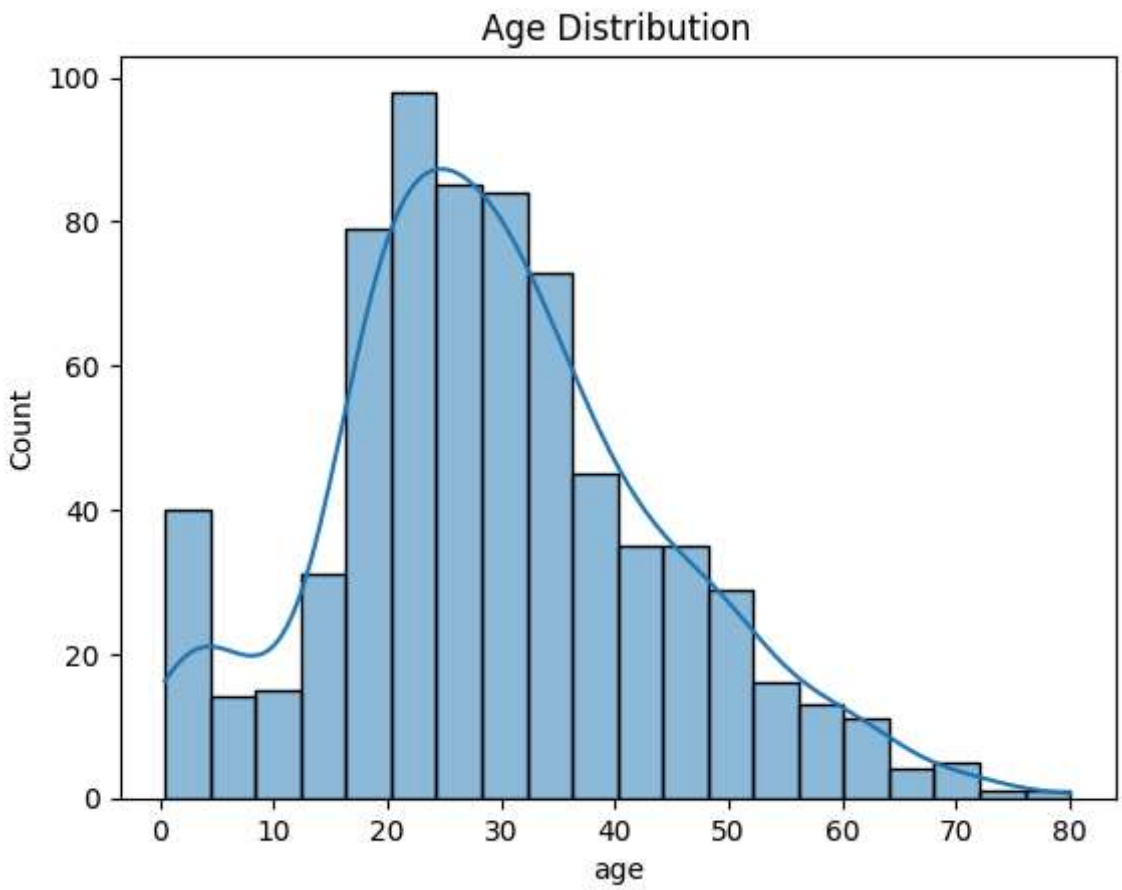
In [6]:
```
#Survival by sex
sns.countplot(x='survived', hue='sex', data=data)
plt.title("Survival by Gender")
plt.xlabel("Outcome")
plt.show()
```



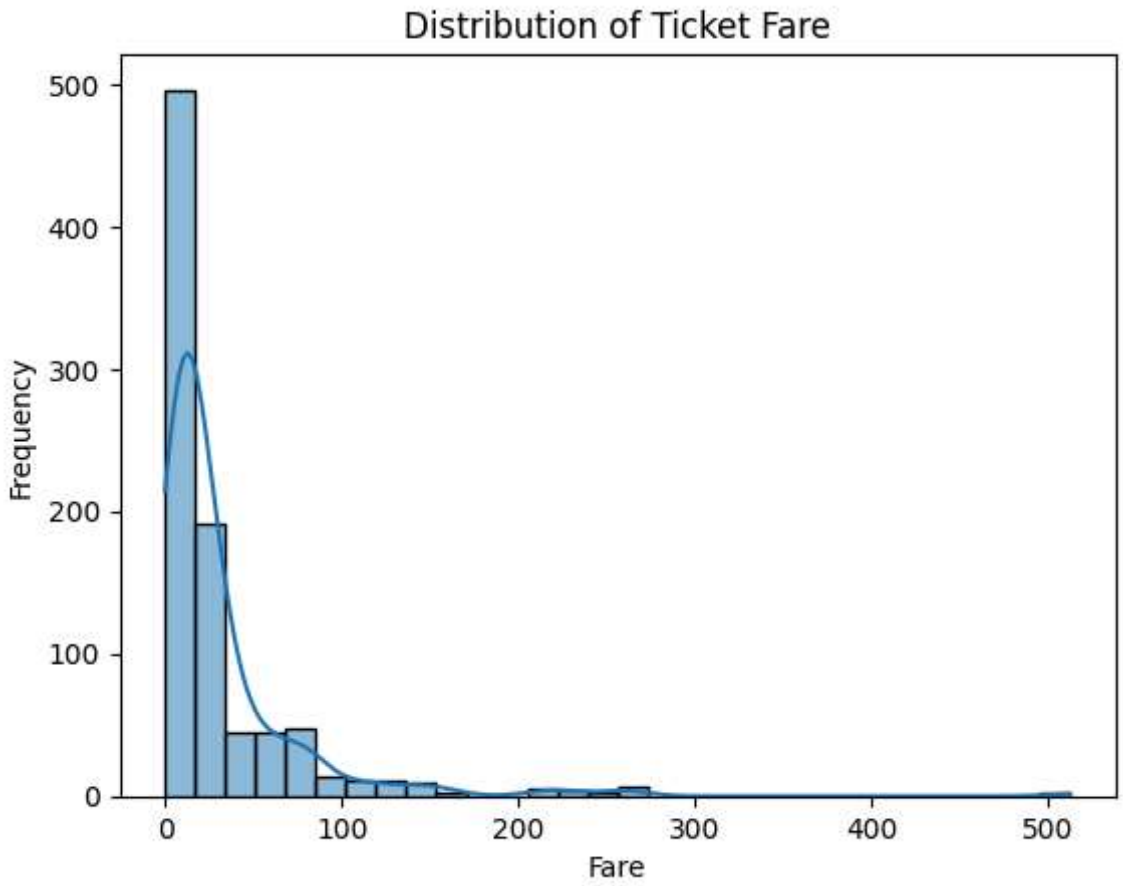
In [7]:
```
#using passenger class
sns.countplot(x='survived', hue='pclass', data=data)
plt.title("Survival by Passenger Class")
plt.xlabel("Outcome")
plt.show()
```



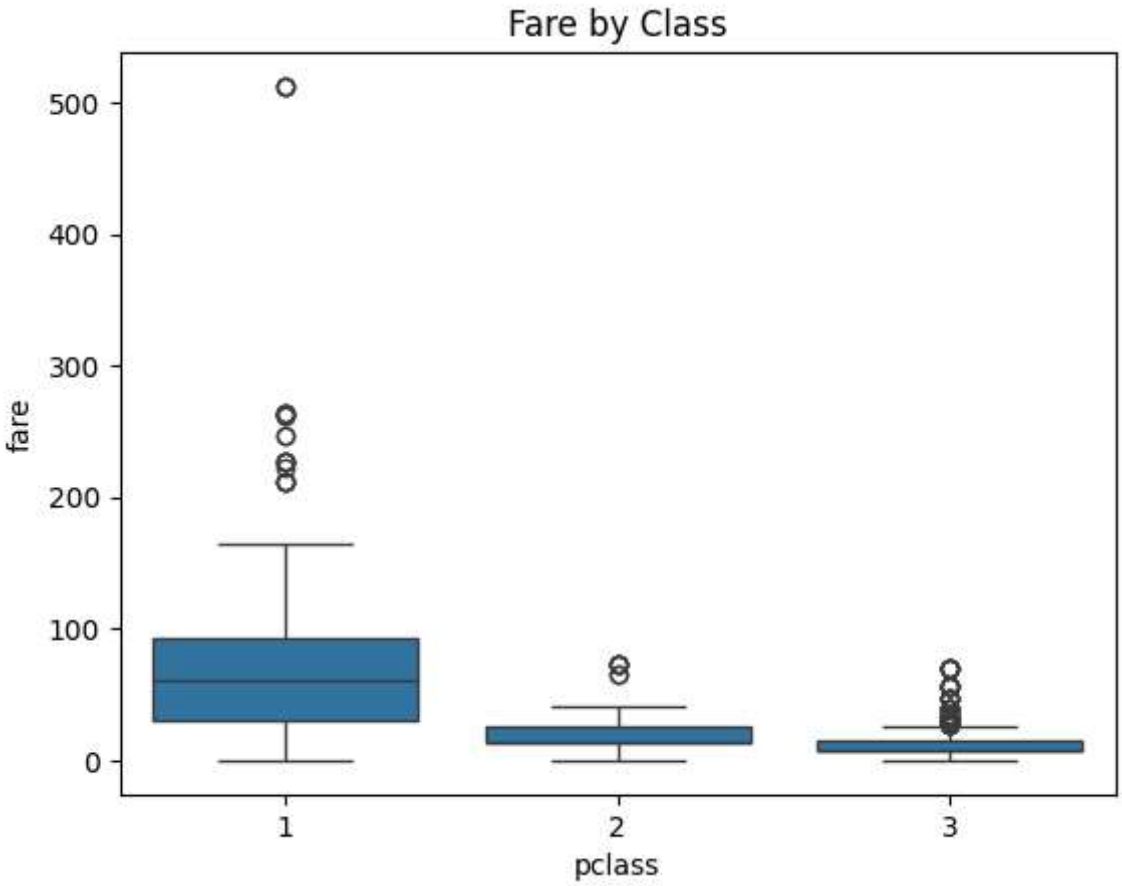
In [8]:
```
sns.histplot(data['age'], kde=True)
plt.title("Age Distribution")
plt.show()
```



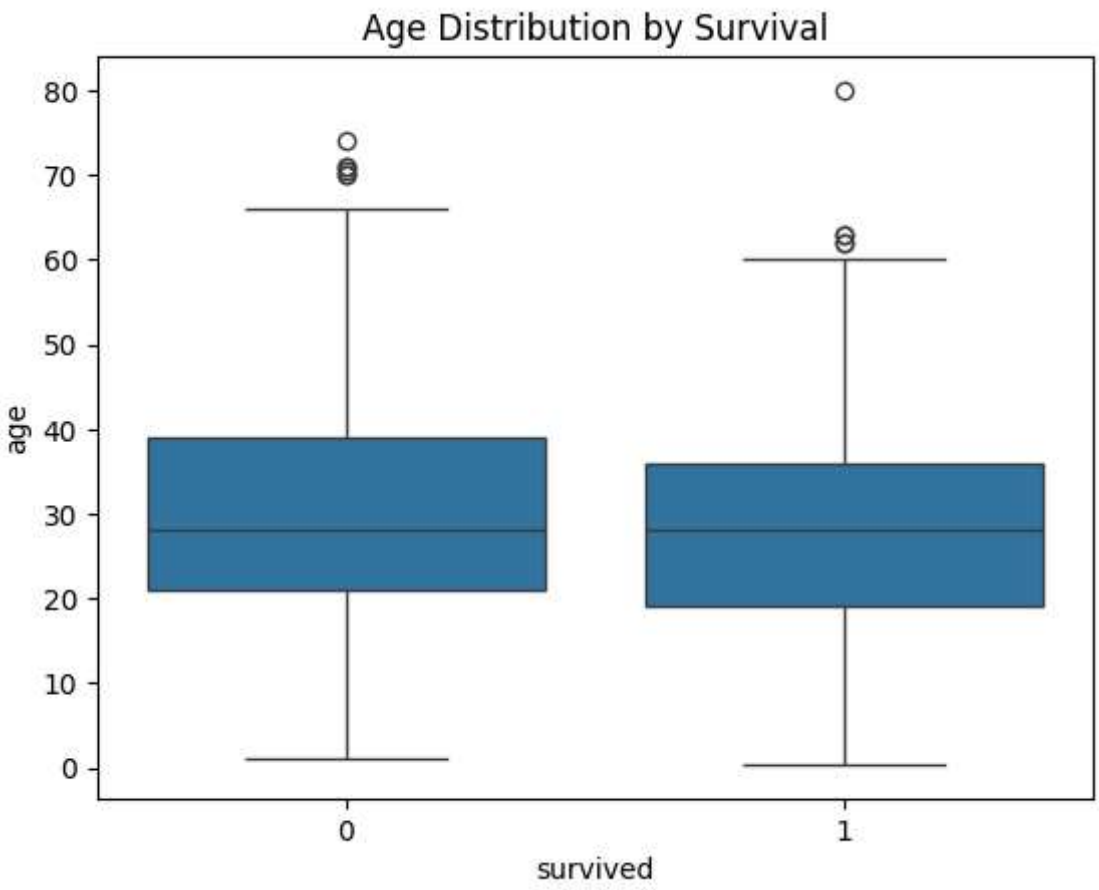
In [9]:
```
#fare distribution
sns.histplot(data['fare'], bins=30, kde=True)
plt.title("Distribution of Ticket Fare")
plt.xlabel("Fare")
plt.ylabel("Frequency")
plt.show()
```



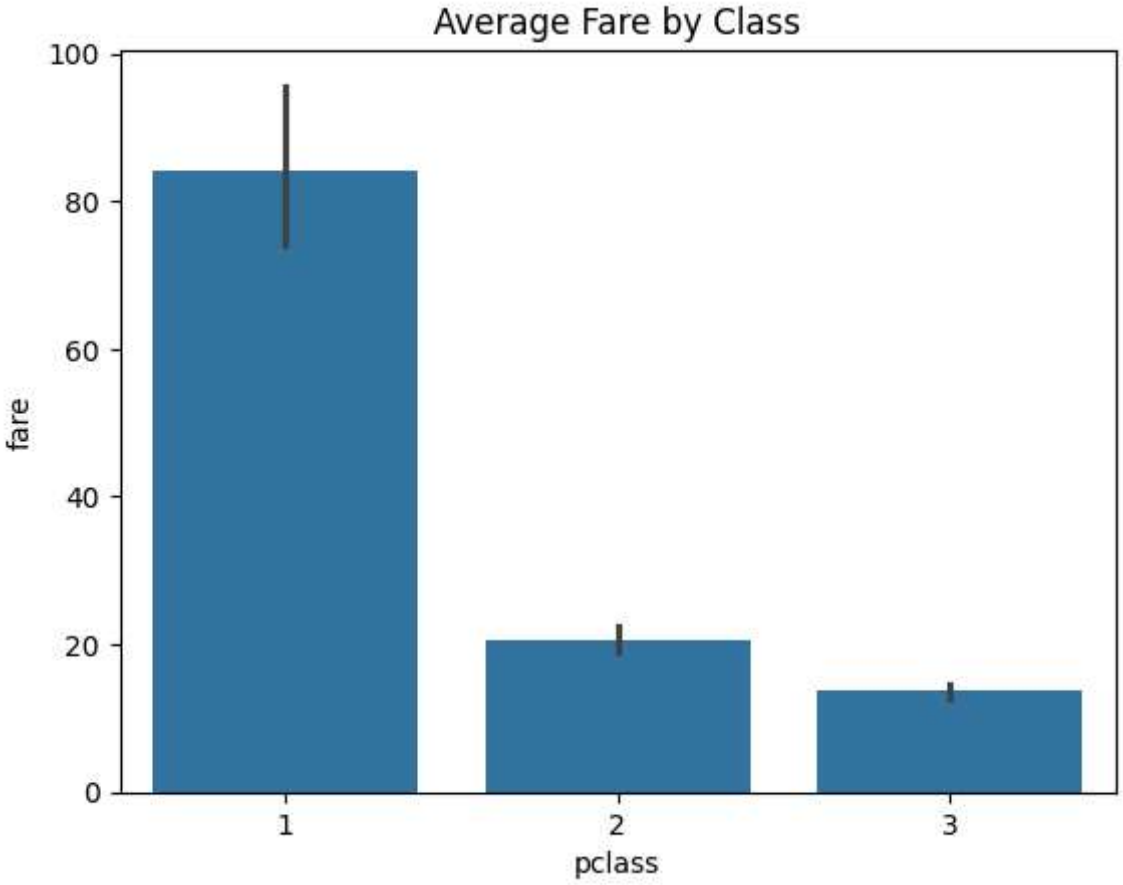
In [10]:
```
sns.boxplot(x='pclass', y='fare', data=data)
plt.title("Fare by Class")
plt.show()
```



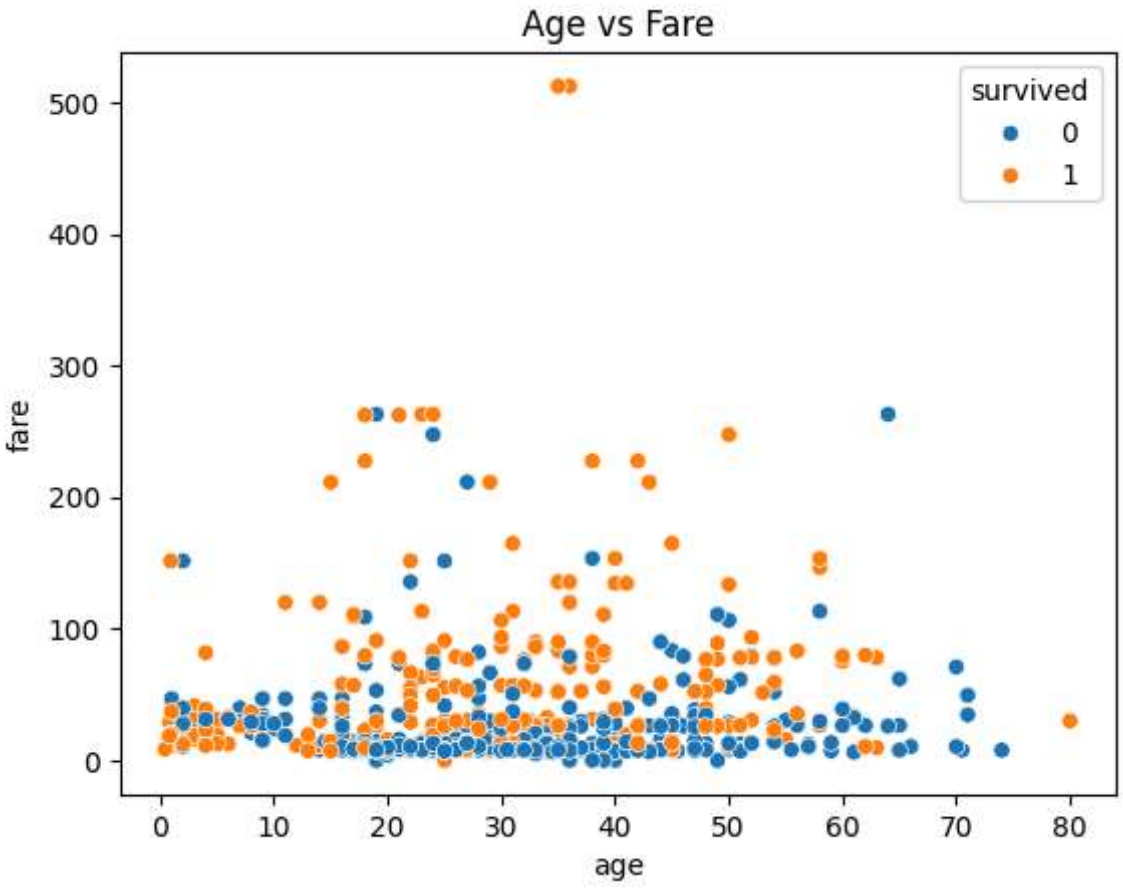
In [11]:
```
sns.boxplot(x='survived', y='age', data=data)
plt.title("Age Distribution by Survival")
plt.show()
```



In [12]:
```
sns.barplot(x='pclass', y='fare', data=data)
plt.title("Average Fare by Class")
plt.show()
```



In [13]:
```
sns.scatterplot(x='age', y='fare', hue='survived', data=data)
plt.title("Age vs Fare")
plt.show()
```


Age vs Fare
survived
0
1
fare
500
400
300
200
100
0
0
10
20
30
40
50
60
70
80
age

In [ ]: